

Rajalaxmi Rajagopalan

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Education

University of Illinois, Urbana-Champaign

PhD, ECE. *Expected Graduation: May 2026*

Relevant Coursework - Optimization, vector space signal processing/optimization, Random processes, Machine Learning for Signal Processing, Information Theory, Statistical Inference

Fall 2021 - Present

GPA 4.0

National university of Singapore

Master of Science

Department of Electrical & Computer Engineering

Fall 2018 - Spring 2020

GPA 4.45/5

CEG, Anna University, India

Bachelors in Engineering

Department of Electrical & Computer Engineering

Fall 2012 - Spring 2016

GPA 9.75/10

Publications

- [1] **R. Rajagopalan**, R. Giri, Z. Tang, K. Han, and G. Friedland, "Masked Autoencoders as Universal Speech Enhancer", ICASSP, April 2025.
- [2] **R. Rajagopalan**, Y.L. Wei, and R.R. Choudhury, "Kernel Learning for Sample Constrained Black-Box Optimization", AAAI, February 2025.
- [3] **R. Rajagopalan**, Y.L. Wei, and R.R. Choudhury, "Sample-Constrained Black Box Optimization for Audio Personalization", AAAI, February 2024.
- [4] **R. Rajagopalan**, Y.L. Wei, and R.R. Choudhury, "Audio Personalization through Human-in-the-loop Optimization", NeurIPS Workshop on ML for Audio, December 2023.
- [5] Y.L. Wei, **R. Rajagopalan**, B. Islam, and R.R. Choudhury, "Self-supervised Speech Enhancement using Multi-Modal Data", NeurIPS Workshop on ML for Audio, December 2023.
- [6] **R. Rajagopalan** and C. K. Tham, "Active Learning for IoT Data Prioritization in Edge Nodes Over Wireless Networks," IECON 2020 The 46th Annual Conference of the IEEE Industrial Electronics Society, Singapore, 2020.

Honors and Awards

■ A.R. "Buck" Knight Fellowship	UIUC	Fall 2025
■ Amazon-Illinois (AICE) Fellowship	UIUC	Spring 2025
■ Exceptional Academics Award	National University of Singapore	Spring 2020
■ Silver Medal- Department second rank award	CEG, Anna University	Spring 2016
■ Gold Medal- Excellence in Mathematics	CEG, Anna University	Spring 2013

Research Experience

Graduate Student Researcher

Signals & Inference Research Group (SiNRG)

Fall 2021 - Present

UIUC

- Sample Constrained Black-Box Optimization using techniques like **Bayesian Optimization Gaussian Process Regression** for Human-in-the-loop personalization of content: audio, images, etc.
- Multi-modal universal speech enhancement through self-supervised learning.
- Other projects in acoustic signal processing and earable computing.

Graduate Student Researcher

Wireless Networking Group

Fall 2018 - Spring 2021

NUS, Singapore

- **Distributed modeling** and **Bayesian statistical inference** in resource-constrained large-scale distributed parallel computing areas: wireless sensor networks and IoT.

Undergraduate Student Researcher

Integrated Systems Lab (ISL)

Spring 2014 - Spring 2016

CEG, India

- ANUSAT-2 satellite (QB50 European Research Mission)
<https://www.qb50.eu/index.php/news.html>

Work Experience

Applied Scientist Intern

AWS AI Labs

Summer 2024, 2025

- Speech and audio-related problems in the Transcribe AWS team.
- Self-Supervised Universal Speech Enhancement.

- Music Generation via Inference-time Optimization of Diffusion Models.

Teaching Assistant
UIUC

Spring 2023

Research Assistant
SiNRG, UIUC

Fall 2021 - Present

- Conducting research in audio signal processing and black-box optimization.

Teaching Instructor
NUS

Spring 2020 - Spring 2021
Singapore

Network Planner - Assistant manager
Airtel

June 2016 - July 2018
Gurgaon, India

Skills

- **Languages :** Python, R, C, C++, VHDL, MATLAB, VBA
- **Hardware Platforms :** USRP, Raspberry PI, BlackFin BF609, BladeRF, SDR, Cyclone –II FPGA, Stratix-III FPGA, TIVA.
- **Certifications:** Cisco Network Master
- **Languages:** English, Tamil, Hindi, French, Japanese
- **Extracurriculars:** Trained Indian classical violinist

Selected Projects

Sample Constrained Black-Box Optimization for Personalization

2021-Present

Prof Dr. Romit Roy Choudhury - SiNRG, UIUC

- We consider the problem of personalizing content to a user's taste. Content could be audio signals in a hearing aid, customized images, etc. Given a content generator space, our goal is to find the optimal sample that will maximize the user's personal satisfaction. The user satisfaction function is unknown, and hence optimizing it using the human in the loop, while constrained by the number of samples, is a sample-constrained black-box optimization problem. We build on the Bayesian Optimization framework. We achieve sample efficiency by developing kernel learning techniques that learn a unique kernel that models the function structure.

MusicX - Guided Music Generation via Diffusion Inference-time Optimization

Summer 2025

Turab Iqbal, Erfan Soltanmohammadi - AWS AI Labs

- We propose MusicX, a training-free, fast, and versatile music generator that modifies music generated by a generic frozen text-to-audio latent diffusion model to generate new music with specific characteristics that the frozen model is incapable of by, (1) Guidance Control: that modifies the diffusion trajectories through an optimization in the diffusion latent space, enabling training-free modification for arbitrary conditioning modalities. (2) Fused Diffusion: merges multiple guided diffusion trajectories, enabling variable-length multi-conditioned generation. (3) Fast Distill: speeds up the distillation by 1000x by modifying the CTM distillation scheme.

Self-supervised Universal Speech Enhancement

Summer 2024

Ritwik Giri, Zhiqiang Tang - AWS AI Labs

- Speech is often corrupted by many distortions like ambient noise, reverberation, limited bandwidth, codec artifacts, interfering speakers etc. In practical scenarios, there is lack of high-quality clean speech references. Thus, we developed a universal speech enhancer that (1) can tackle a broad range of distortions simultaneously, (2) trained in a self-supervised manner without access to large-scale clean speech corpus, and (3) offers robust performance for enhancement downstream tasks.

Self Supervised IMU-based Speech Enhancement

2021-Present

Prof Dr. Romit Roy Choudhury - SiNRG, UIUC

- Extracting compact representations of personalized speech features like base frequencies using representation models like autoencoders from captured surface vibrations over the face from the speakers' throat by IMUs in earphones, to perform personalized denoising of speech corrupted by over-the-air interference.

Master's Thesis: Active Learning for IoT Data Prioritization in Edge nodes over Wireless Networks

2019

Prof Dr. Chen-Khong Tham - WICOMM, NUS

- Developed Active learning-based and Bayesian ML-based techniques to improve the robustness of distributed deep learning models in the presence of practical wireless communication channel issues for the application of predictive maintenance of industrial machines.

Design of Anusat-2 Nano-Satellite (QB50 European Research Mission)

2015

Prof Dr. P. V. Ramakrishna - ISL, CEG

- Designed end-to-end protocol stack including the telecommand and telemetry packet structure for communication with the OBC (On-Board Computer) and FIPEX payload of the satellite and ground station.
- Designed the MCD (Mission control data center) that creates a repository of all payload data, periodic housekeeping data, and Whole Orbit Data, runs scheduled scripts to send periodic commands to the satellite for monitoring the status of various modules.
- Calibration and testing of the ADCS (Attitude Detection and Control System) responsible for stabilizing the satellite in its orbit. Coding the OBC to control the various modules like ADCS, FIPEX payload, solar panels, comms system, etc., and error handling procedures.